

Probing the trivial vacuum of the BMN model

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October 26, 2020

Abstract

The matrix model on a plane-wave background is discussed in the classical regime. We propose a geometry for the trivial vacuum motivated by a specific limit that arises from the M-theory description where the data of M-theory can be mapped to an electrostatic problem in three dimensions. Furthermore, we investigate the trivial vacuum of the BMN model and give a gravity description in terms of D0-branes forming and probing this geometry. Using a single D0-brane as probe, we find a repulsive behavior approaching the core of this geometry. We comment on implications believing also that it lies in the reach of Monte-Carlo simulations.

1 Introduction

Matrix models have played a significant role in the development of quantum gravity theories. In a sense it seems that they can lead to emergent (quantum) gravity theories [?], [?], [?], [?]. In this work we focus on matrix models motivated by M-theory [?], [?]. These models are believed to give a non-perturbative *definition* of string and M-theory. In particular, it seems that they capture more characteristics of quantum gravity than string theory, and for this reason they are believed to be more fundamental than string theory itself [?].

Even though there is a plethora of these, here we focus in the BMN model [?]. Before dive in this particular model let us explain the main idea of these models. One starts from a two-sided approach, one of which is M-theory and the other is Yang-Mills theory. M-theory naturally lives in a eleven-dimensional spacetime, while on the other hand Yang-Mills theories can be consistently written in any dimension. Accordingly one can write a ten-dimensional analogue of Yang-Mills. Starting from the gauge theory side, the recipe indicates to compactify nine out of ten dimensions of the Yang-Mills theory, such that in the end one is left with just one dimension which is nothing more than $(0+1)$ –quantum mechanics. Since it is somewhat awkward to have one spatial dimension and study dynamics of the system, usually this dimension is connected to time. The variables in this model are Hermitian matrices X^i where $i = 1, \dots, 9$. In a sense, the gauge fields living in the ten-dimensional spacetime upon compactification they become scalar fields in the low dimensional spacetime. Then for each compactified dimension there is an index i for the scalar field. This is the Yang-Mills side.

From the M-theory perspective we have the following procedure: we compactify just one dimension out of eleven which we might take to be a spatial circle of radius R_s . It is understood from the very early work of Kaluza [?] and Klein [?], that compactification on a spatial dimension on a circle gives us for free some modes, the so-called Kaluza-Klein modes. These, appear in the lower dimensional spacetime accompanied with a mass and physically correspond to (longitudinal) momenta running in the compactification circle given by the following relation

$$p = \frac{N}{R_s},$$

where N is the number of modes. In a rough sense one can think of this procedure, as a mechanism that produces mass in a lower dimensional spacetime by compactifying massless modes in spatial dimensions of a higher dimensional spacetime. In our case, one unit of mass is $\frac{1}{R_s}$.

In addition, one has to perform a boost with boosting parameter

$$\gamma = \sqrt{\frac{R^2}{2R_s} + 1}, \tag{1.1}$$

and take the limit $R_s \rightarrow 0$. Physically, this means that we boost along the compactification circle whose radius goes to zero. The result of this is that γ becomes infinite corresponding to a lightlike circle of radius R [?]. This procedure indicates a duality between boosted spatial compactifications with $R_s \rightarrow 0$ and lightlike compactifications. However, in this way we generate closed lightlike curves. At the end of the day to talk about physical implications, one has to take the $R \rightarrow \infty$ limit. In other words, the physically relevant limit is

$$p = \frac{N}{R} = \text{fixed} \quad , \quad N \rightarrow \infty \quad , \quad R \rightarrow \infty. \tag{1.2}$$

Furthermore, we remind that the compactified M-theory along the circle with R_s radius, where $R_s = \sqrt{\alpha'} g_s$ becomes ten dimensional in the $R_s \rightarrow 0$ limit. In fact this is believed to be the type IIA string theory and actually it is used as the definition of it. Moreover, in this limit the low energy sector which consists of D0-branes decouples. One way to realize this is that M-theory contains also membranes and these in general can wrap the compactified circle, however in the boosted frame these membranes become non-local while the longitudinal modes remain local. From the type IIA point of view the only local elements are the D0-branes. Thus, this sector decouples from the rest of the system. The N -th Kaluza-Klein mode corresponds to N units of momentum $\frac{1}{R_s}$ in the compact dimension, and in this case, the $U(1)$ sector is the R-R one form of type IIA. Hence, we interpret D0-charge as eleven-dimensional momentum. In this way the arbitrary Kaluza-Klein modes on the compactified circle we described above are massless modes and are materialized as supergravitons. Being themselves massless in the eleven-dimensional spacetime, they appear in the ten-dimensional spacetime as pointlike D0-branes. Having sketch the connection of D0-branes with supergravitons (of eleven-dimensional M-theory), one might ask what about the rest of the objects like membrane or fivebrane? As it is nicely explained in [?], there is no need to add extra degrees of freedom in the action to describe also these objects, but rather they can be described via D0-branes and T-dualities.

The procedure of boosting, relates the non-relativistic ten dimensional theory of many D0-branes (large N) with a particular frame of the M-theory, the so-called infinite momentum frame (IMF). Then the conjecture is that the dynamics of the D0-branes are in one-to-one correspondence with the Hermitian matrices of the $(0+1)$ -dimensional Yang-Mills theory and they describe all the contents of M-theory in this particular IMF frame¹. A graphical representation of the above is given in figure 1. The previous statement holds when N is large. Further investigations have proposed that this should be true even at finite N [?]. However since the equivalence principle breaks down for the finite N version of the theory [?], [?] it is quite unlikely that it can be related to a smooth theory of Einstein-Hilbert gravity. On the other hand, this might not be a real problem since in the spirit of holography, finite N corrections seem to correspond to quantum corrections, and from many different approaches quantum spacetime is believed to be discrete. Thus, it is more likely that the finite N version of the model would correspond to a (quantum corrected) gravity theory on a non-commutative space.

Another realization of this duality which might be seen as evidence is the following. Firstly, there is lack of a concrete microscopic description of M-theory, and since the contents of M-theory are supermembranes, the UV completion of this theory has to be regularized via these membranes. This procedure [?] results in a Hamiltonian containing bosonic (X 's) and fermionic (Ψ 's) terms

$$H_0 = R \text{Tr} \left(\frac{1}{2} \dot{X}_A \dot{X}_A - \frac{1}{4} [X_A, X_B]^2 - \frac{1}{2} \Psi^T \gamma^A [X_a, \Psi] \right), \quad A, B = 1, \dots, 9. \quad (1.3)$$

In the same Hamiltonian, one can arrive by the dimensional reduction of the $U(N)$ 10D-SYM theory. Also, since the strong coupling limit of type IIA string theory is supposed to be M-theory, one should also arrive in (1.3) by using type IIA terminology. Indeed, this is the case for a large number of D0-branes at low energy and weak coupling [?], [?]. Thus, we see that one can reach this particular Hamiltonian (1.3) by three different approaches, M-theory, type IIA string theory and SYM theory, and this strongly suggests that there must be an underlying relation among them. This was one of the motivations of Banks, Fischler, Shenker, Susskind to propose their conjecture [?].

¹Note however that the M5-brane of M-theory can not be easily realized from the Matrix model [?].

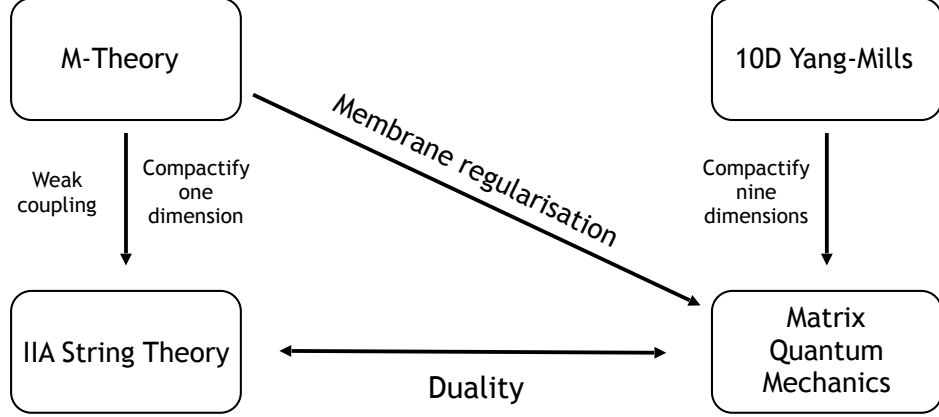


Figure 1: The correspondence between supersymmetric matrix quantum mechanics and M-theory. In this limit the type IIA string theory is in a specific frame, the infinite momentum frame ($N \rightarrow \infty$).

It was only later when another proposal [?] (BMN matrix model), realized also [?] as a special case of a one parameter (μ) family of maximally supersymmetric (32 supercharges) matrix models, whose Hamiltonian is a deformation of (1.3) by

$$H = H_0 + \frac{R}{2} \text{Tr} \left(\sum_{i=1}^3 \left(\frac{\mu}{3R} \right)^2 X_i^2 + \sum_{a=4}^9 \left(\frac{\mu}{6R} \right)^2 X_a^2 + i \frac{2\mu}{3R} \epsilon^{ijk} X_i X_j X_k + i \frac{\mu}{4R} \Psi^T \gamma^{123} \Psi \right). \quad (1.4)$$

The original proposal [?] for general flux μ was supposed to give a description of the Discrete Light Cone Quantization (DLCQ) of M-theory on the maximally supersymmetric pp-wave background² of eleven dimensional supergravity

$$ds^2 = -2dx^+ dx^- + \sum_{A=1}^9 dx^A dx^A - \left(\sum_{i=1}^3 \frac{\mu^2}{9} x^i x^i + \sum_{a=4}^9 \frac{\mu^2}{36} x^a x^a \right) dx^+ dx^+, \quad (1.5)$$

with

$$F_{123+} = \mu \quad , \quad x^\pm = \frac{t \pm x}{\sqrt{2}}. \quad (1.6)$$

In the large N limit with fixed momentum $p = \frac{N}{R}$ and fixed μ , this matrix quantum mechanics model should describe M-theory on the pp-wave (1.5) background in a sector with boost-invariant momentum parameter μp .

²For general plane-wave backgrounds see [?].

We can write the bosonic part of the potential of (1.4) in sum of positive terms such as

$$V = \frac{R}{2} \text{Tr} \left[\left(\frac{\mu}{3R} X^i + i\epsilon_{ijk} X^j X^k \right)^2 + \left(\frac{\mu}{6R} \right)^2 X_a X^a \right]. \quad (1.7)$$

Then the types of vacua of this theory are manifest. They are

- $X^a = 0 = X^i$, $i = 1, 2, 3$, $a = 4, \dots, 6$ and corresponds to the trivial vacuum
- $X^a = 0$, $a = 4, \dots, 9$ and $X^i = \frac{\mu}{3R} J^i$, $i = 1, 2, 3$, with J^i being the generators of $SU(2)$ satisfying $[J_a, J_b] = i\epsilon_{abc} J_c$ and corresponds to the non-trivial vacuum.

Among the quantum mechanics data and the gravity data there is only one relationship

$$\text{matrix elements } X \longleftrightarrow \text{D0-branes}, \quad (1.8)$$

and this is sufficient to reconstruct all the contents of gravity. The last statement has to do with the remarkable properties of D0-branes, which in the presence of suitable fluxes they can polarize into D2- and NS5-branes. Then studying only X amounts to study all the contents of the gravity theory in the following way. Each diagonal element of X corresponds to the position of each D0-brane in the gravity side, and the non-diagonal elements to stringy interactions among the D0's, see figure 2. This is understood in the following way. Being the matrices X (sub)part of the Yang-Mills theory³ they also transform under the adjoint representation of $U(N)$ symmetry because of unitarity. We recall that under the same representation transforms an open string. Hence, both for the matrix and the string we need two indices to label the "state", which from the matrix point of view these indices can be realized as indices labelling the entries of the matrix, while from string perspective these are the Chan-Paton indices. Hence, for instance a string stretching from D-brane 1 to D-brane 2 is labelled as a state $|12\rangle$ and corresponds to the x_{12} entry of the matrix. Similarly, strings ending on the same D-brane are denoted with same indices, but since D0-branes are point-like the information we get is that this open string actually begins and ends on the same point. Then it is understood that since the size of the matrix is $N \times N$ we have to take N large such that we reconstruct more and more information about the gravity data.

Furthermore, the BMN matrix model has one more scale given by the mass μ . Then one can construct a dimensionless, effective coupling as $g_{\text{eff}} = \frac{\lambda}{\mu^3}$, where λ is the t'Hooft coupling $\lambda = g_{YM}^2 N$. This effective coupling controls the regimes of validity of the descriptions. Weak coupling ($g_{\text{eff}} \ll 1$) is when the matrix theory is valid and perturbation analysis can be applied with g_{eff} as expansion parameter. Strong coupling ($g_{\text{eff}} \gg 1$) is when the supergravity description is valid. In this paper we are interested in the gravity description for the bosonic sector of the model.

In addition we comment to a naive interpretation of the solutions of the BMN potential (1.7) from the gravity point of view. The trivial vacuum corresponds to placing N D0-branes in $r = 0$ in the gravity side creating in this way a black hole-like object in type IIA supergravity with $SO(3) \times SO(6)$ symmetry. In the BFSS model this is easier to realize since the metric of N coincident D0-branes has naturally a $\mathbb{R} \times SO(8)$ symmetry. For the BMN model instead things are not so easy since we should realize a metric that has $\mathbb{R} \times SO(6) \times SO(3)$ symmetry. From M-theory point of view, the

³In particular it is dimensional reduction of it as we saw.

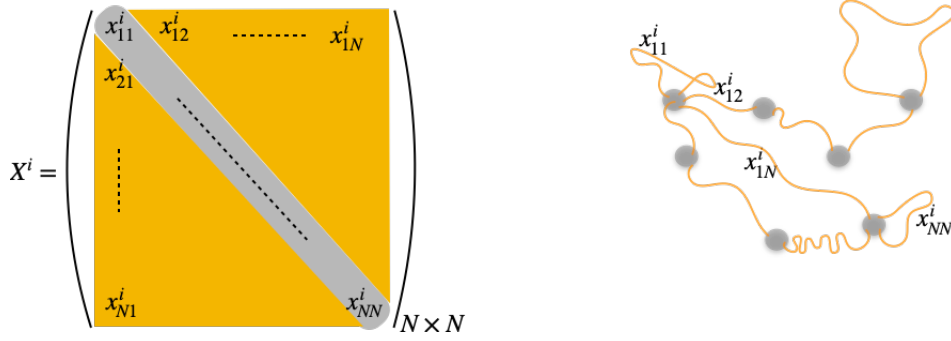


Figure 2: The correspondence between the matrices of quantum mechanics and D0-branes of string theory. Diagonal elements contain information for the positions of D0's while non-diagonal elements indicate interactions between them. The D0-branes are point-like, hence an entry $x_{\alpha\alpha}$ gives the "location" of the α -th D0-brane.

trivial vacuum is believed to be due to a single 5-brane [?]. On the other hand, the number of non-trivial vacua can be organized in partitions of N as $\{N_1, \dots, N_k\}$. Geometrically they are supposed to correspond to concentric fuzzy shells in three dimensions placed at the center of the transverse six dimensions, with each shell having radius analogous to it's representation and the flux parameter μ , i.e $r_i \sim \frac{N_i \mu}{6R}$ [?],[?].

Nevertheless there has being a lot of progress towards understanding of the matrix models in the context of (quantum) gravity [?], [?], [?], [?], [?].

In this paper, we have organized the contents as follows. In the next section we investigate the dynamical behavior of D0-branes in a specific background, in particular in a symmetry reduced, classical pp-wave background. We do this by firstly calculating the force that a single (probe) D0-brane feels once it has been inserted in this background and secondly by calculating the force a probe D2-brane feels in the same background. One can think that this D2-brane can be realized as consisting of a large number ($q \gg 1$) of D0-branes taking into account a specific limit (see section 2.2). Furthermore, in section 3 after discussing the extremal limit of the geometry we repeat the analysis for a D0-brane in a background with a specific limit that geometrically corresponds to the trivial vacuum solution of the BMN at finite temperature. We then conclude with some remarks in section 4.

2 Force in symmetry reduced pp-waves: a warm up

In this section we start by calculating the force that a D0-brane feels in a classical plane wave background in ten dimensions and then repeating the calculation for a D2-brane. The starting point is an eleven dimensional metric which is a symmetry reduced version of the (1.5)

$$ds^2 = -(1 + \frac{Q^2}{4})dt^2 + (1 - \frac{Q^2}{4})(dx^{10})^2 - \frac{Q^2}{2}dtdx^{10} + \sum_{i=1}^3 dx_i dx_i + \sum_{a=4}^9 dx_a dx_a, \quad (2.1)$$

where Q is defined as ⁴

$$Q^2 := \frac{\mu^2}{9} \sum_{i=1}^3 (X^i)^2 + \frac{\mu^2}{36} \sum_{a=4}^9 (X^a)^2, \quad (2.2)$$

⁴This μ is the deformation parameter, and it is the same that appears in the matrix model (1.4).

and the 4-form field is given as

$$Q_{123(10)} = -Q_{0123} = \frac{\mu}{2}. \quad (2.3)$$

We note that this metric, is written in the unboosted frame. Since any pp-wave metric is boost covariant [?], the effect of boosting is simply to rescale the deformation parameter μ . Furthermore, we note that this background is a general background that should capture general (but symmetry reduced) features of the BMN matrix model and in fact one can derive the action of the BMN from this metric [?], [?].

2.1 A D0-brane feels only the trivial vacua

We want to perform a compactification of (2.1) along the spacelike x^{10} direction to make contact with the ten-dimensional theory. One easy way to obtain this is to write down the relation between the 11D-metric and the 10D-metric

$$ds_{11}^2 = e^{-\frac{2\phi}{3}} ds_{10}^2 + e^{\frac{4\phi}{3}} (dx^{10} - C_\alpha dx^\alpha)^2. \quad (2.4)$$

Then, the ten-dimensional metric is given as

$$ds_{10}^2 = - \left(1 - \frac{Q^2}{4}\right)^{-\frac{1}{2}} dt^2 + \delta_{ab} \left(1 - \frac{Q^2}{4}\right)^{\frac{1}{2}} dx^a dx^b, \quad (2.5)$$

with

$$e^{\frac{4\phi}{3}} = 1 - \frac{Q^2}{4}, \quad C_0 = -\frac{Q^2}{4 - Q^2} dt, \quad Q_{0123} = -\frac{\mu}{2}, \quad H_{123} = \frac{\mu}{2}. \quad (2.6)$$

Here C_0 is the RR-one form potential, H_{123} is the NS-NS three-form potential and F_{0123} is the RR three-form field strength. These form fields arise from the distribution of the eleven-dimensional three form into the R-R and NS-NS sector. Geometrically, this corresponds to distributing the eleven dimensional momentum into different geometric configurations of D-branes. When having in mind a single D0-brane we will consider only the one-form since it is the one that the D0-brane is naturally coupled with.

By direct calculation of the Ricci scalar of (2.5), we find

$$\begin{aligned} \mathcal{R} = \frac{\mu^2}{5184} \left(1 - \frac{Q^2}{4}\right)^{-\frac{5}{2}} \sum_{i=1}^3 \sum_{a=4}^9 \left\{ \mu^2(7 - Q^2) \left[(X^a)^4 + 48(X^a)^2(X^i + 3) + 96X^a(X^i)^2 \right. \right. \\ \left. \left. + 16(X^i)^2((X^i)^2 + 36) \right] \right. \\ \left. + 252[(X^a)^2 + 4((X^i)^2 + 9)]Q - 63[(X^a)^2 + 4((X^i)^2 + 9)]Q^3 \right\} \end{aligned} \quad (2.7)$$

From this we see that we can trust the supergravity solution (2.5) only in the low curvature region, i.e when $\frac{Q^2}{4} \ll 1$.

In this limit one can expand as follows

$$e^{-\frac{2\phi}{3}} \approx 1 + \frac{Q^2}{8} + \mathcal{O}(Q^4)$$

$$C_0 \approx -\frac{Q^2}{4} + \mathcal{O}(Q^4).$$

Having set up the playground, now we turn to calculate the force. When we do this, we have in mind a situation when we naively place a D0-brane probe in the spacetime and ask for its dynamical behaviour. To this end, we need to write down the DBI action for the D0-brane by choosing the gauge where the parameter of the worldline of the D0 coincides with the time component of the spacetime. In a general background, it is given by

$$S_{D_0} = -T_0 \int dt e^{-\phi} \sqrt{-g_{\mu\nu} \frac{dx^\mu}{dt} \frac{dx^\nu}{dt}} + T_0 \int dt C_0, \quad (2.8)$$

where $T_0 = \frac{M_s}{g_s}$ is the tension of the D0-brane and the expression inside the square root is the pull-back of the spacetime metric on the worldline of the D0-brane. We remind that a Dp-brane naturally couples with a p+1 form, thus the only form that couples (naturally) to a single D0-brane is the one-form ⁵ potential C_0 . Then, the action reads as

$$S_{D_0} = -T_0 \int dt \left(1 + \frac{3Q^2}{16}\right) \sqrt{1 + \frac{Q^2}{8} - \left(1 - \frac{Q^2}{8}\right) \dot{x}^2} + T_0 \int dt \left(-\frac{Q^2}{4}\right). \quad (2.9)$$

Moreover, after taking the non-relativistic limit we expand the action up to $\dot{x}^2$ and Q^2 terms leading to

$$S_{D_0} \approx -T_0 \int dt \left[1 + \frac{Q^2}{2} - \frac{1}{2} \dot{x}^2\right] + \mathcal{O}(Q^4, \dot{x}^4). \quad (2.10)$$

One can read off the potential from this action:

$$V_{D_0}(x) = T_0 \left(1 + \frac{Q^2}{2}\right). \quad (2.11)$$

Recalling (2.2) we write the potential as,

$$V_{D_0}(X^i, X^a) = T_0 \left[1 + \frac{1}{2} \left(\frac{\mu^2}{9} \sum_{i=1}^3 (X^i)^2 + \frac{\mu^2}{36} \sum_{a=4}^9 (X^a)^2 \right)\right]. \quad (2.12)$$

We see an explicit dependence of μ in the potential. It is also worth noting that the trivial vacua ($X^i = 0$ and $X^a = 0$) are supersymmetric solutions of the potential after performing a shift, but the non-trivial $SU(2)$ vacua are not. The latter, geometrically are believed to correspond to concentric fuzzy spheres, however the single D0-brane in this background is agnostic if there are any fuzzy spheres at all. Had we consider a different probe, for example a D2-brane, then this would couple to higher forms and would give rise to the $SU(2)$ part of the potential. This is demonstrated in the next subsection.

⁵We could of course consider a three-form flux and via this procedure the D0-brane would couple to also a three form, giving rise to a Myers effect. This would change the output of the discussion but we change in the first place the background by the introduction of the three-form.

Furthermore, we can write the terms inside the sum as the radii of an S^2 and S^5 sphere. In other words we can write the potential as

$$V_{D_0}(r_2, r_5) = T_0 \left[1 + \frac{1}{2} \left(\frac{\mu^2}{9} r_2^2 + \frac{\mu^2}{36} r_5^2 \right) \right], \quad (2.13)$$

where r_2 denotes the radius of the S^2 sphere and r_5 the one of S^5 .

From this potential one can obtain the force that the D0-brane feels by differentiating with respect to the radial coordinates. This is simply given by

$$F_{D_0}(r_2, r_5) = -T_0 \frac{\mu^2}{9} \left(r_2 + \frac{r_5}{4} \right). \quad (2.14)$$

However, this is a naive picture because the hole system is studied at the classical level. According to matrix model simulations, the matrices are never zero but they subject to quantum fluctuations independent of which vacuum one studies [?], [?], [?], [?].

2.2 A D2-brane feels the $SU(2)$ vacua

In this section we calculate the force that a D2-brane probe feels in the weakly curved background (2.5). One can think this D2 probe as being made of (q) D0-branes where the number of D0's is taken to be large. However, in order to consider this as a probe, one has to require that the probe charge q is much less than the background charge N such that we do not take backreaction into account, i.e we are studying the limit $1 \ll q \ll N$. We can imagine this brane to have a S^2 topology embedded in a three dimensional space. In particular, one natural embedding would be along the $SO(3)$ part of the metric where the other six dimensions are transverse to it. For convenience, we place the D2-brane in the center of the the six transverse directions such that $X^a = 0$. Then the DBI and WZ action of the D2-brane will be of the familiar form

$$S_{D_2} = -T_2 \int d^3\sigma e^{-\phi} \sqrt{-\det(G_{ab} + 2\pi\alpha' \mathcal{F}_{ab})} + T_2 \int (C_3 + 2\pi\alpha' \mathcal{F}_2 \wedge C_1), \quad q \ll N \quad (2.15)$$

where $2\pi\alpha' \mathcal{F}_2 = 2\pi\alpha' F_2 - B_2$. As in the case of the D0-brane, we have also chosen the static gauge here, which corresponds to identify the worldvolume dimensions with the three dimensional part of the subspace of the ten-dimensional spacetime. We can choose t, θ, ϕ coordinates parametrizing the S^2 . The radius of S^2 is given by r , while the angles have the usual ranges $\theta \in [0, \pi]$, $\phi \in [0, 2\pi]$. For a compact D2-brane of arbitrary topology the total charge q is quantized and it is given as

$$\int_{S^2} F_2 = 2\pi q \quad \longrightarrow \quad F_2 = \frac{1}{2} q \sin \theta d\theta \wedge d\phi. \quad (2.16)$$

We note that the integer q is also related to the first Chern class of the $U(1)$ bundle at a fixed point in time [?], [?] and moreover, it acts as a source for the R-R vector field and the interpretation of it is the number of D0-branes that are bound on the surface of the D2-brane.

Analogously, the forms are given in the static gauge as

$$C_1 = \left(1 - \frac{Q^2}{4} \right)^{-1} dt \quad (2.17)$$

$$C_3 = \frac{1}{3} \mu dt \wedge S_2 \quad , \quad S_2 = \frac{1}{2} \epsilon_{ijk} x^i \wedge dx^j \wedge dx^k, \quad i, j, k = 1, 2, 3. \quad (2.18)$$

Furthermore, the pullbacks of the metric G_{ab} on the D2-brane are given as

$$\det g_{tt} = - \left(1 - \frac{Q^2}{4}\right)^{-\frac{1}{2}} \quad (2.19)$$

$$\det g_{\perp} = \det(g_{\theta\theta}g_{\phi\phi}) = \left(1 - \frac{Q^2}{4}\right) r^4 \sin^2 \theta. \quad (2.20)$$

From these we have that

$$F_{\theta\phi} = \frac{1}{2} q \sin \theta \quad (2.21)$$

$$F_{ab}F^{ab} = \frac{q^2}{2 \left(1 - \frac{Q^2}{4}\right) r^4} \quad (2.22)$$

$$\det F_2 = \frac{1}{2} F_{ab}F^{ab} \det g_{\perp} = \frac{q^2}{4} \sin^2 \theta. \quad (2.23)$$

The DBI action can be written as

$$S_{DBI} = -T_2 \int d^3\sigma e^{-\phi} \sqrt{-\det g_{tt}} (\det g_{\perp} + 2\pi\alpha' \det F_2)^{\frac{1}{2}}, \quad (2.24)$$

and considering the main contribution coming from $\det F_2$ ⁶, we can expand the action around this and get

$$S_{DBI} \approx -T_2 \int d^3\sigma e^{-\phi} \sqrt{-\det g_{tt}} \left(2\pi\alpha' \sqrt{\det F_2} + \frac{\det g_{\perp}}{4\pi\alpha' \sqrt{\det F_2}} \right). \quad (2.25)$$

Using the exact expressions this gives

$$\begin{aligned} S_{DBI} &\approx -T_2 \int dt d\theta d\phi \left[\left(1 - \frac{Q^2}{4}\right)^{-1} \pi\alpha' q \sin \theta + \frac{r^4 \sin \theta}{2\pi\alpha' q} \right] \\ &\approx -T_2 \int dt \left[\left(1 - \frac{Q^2}{4}\right)^{-1} 4\pi^2 \alpha' q + \frac{2r^4}{\alpha' q} \right]. \end{aligned} \quad (2.26)$$

Similarly the WZ action gives

$$S_{WZ} = T_2 \int dt \left[\frac{4\pi\mu r^3}{3} + 4\pi^2 \alpha' q \left(1 - \frac{Q^2}{4}\right)^{-1} \right]. \quad (2.27)$$

We see that the leading terms cancel each other, the subleading terms that remain can be mapped to the commutator term $\sim \text{Tr} [X^i, X^j]^2$ and the Myers term $\sim -\text{Tr} (i\frac{\mu}{3}\epsilon_{ijk}X^iX^jX^k)$ in the BMN model. In addition, one could also be interested in the contributions coming from the B_2 field. However, as it is shown in [?] the contributions of B_2 coming from WZ and DBI action cancel each other. Nevertheless, we can add a term $\sim \mu^2 r^2$ such that we complete the square in the action. This will make the interpretation more clear. By adding and subtracting this term, we write the action as

$$S_{D_2} \approx -\frac{2T_2}{\alpha' q} \int dt \left[\left(r^2 - \frac{2\pi\mu\alpha' q}{3} r \right)^2 - \frac{4\pi^2 \mu^2 \alpha'^2 q^2 r^2}{9} \right]. \quad (2.28)$$

⁶This amounts to $\frac{q^2}{4} \sin^2 \theta \gg 1$, reflecting the fact that the charge q is large but nevertheless we are in the limit $q \ll N$

The potential is simply

$$V_{D_2}(r) = \frac{2T_2}{\alpha'q} \left[\left(r^2 - \frac{2\pi\mu\alpha'q}{3}r \right)^2 - \frac{4\pi^2\mu^2\alpha'^2q^2r^2}{9} \right]. \quad (2.29)$$

From this we see that apart from the trivial $r = 0$ solution, there is a non-trivial equilibrium radius at

$$r = \frac{4\pi\alpha'q\mu}{3}. \quad (2.30)$$

This is understood as the radius in which the contraction that is caused by the r^2 and r^4 terms compensates with the expansion caused by the r^3 term in the potential. In fact one can generalize this and have a number (k) of concentric D2-branes instead of just one. Then for each D2-brane there is an equilibrium radius given by

$$r_i = \frac{4\pi\alpha'q_i\mu}{3}, \quad i = 1, \dots, k, \quad (2.31)$$

with q_i being the charge of the i -th D2-brane. In the Matrix model side instead, this picture corresponds to a particular copy of the N dimensional representation of $SU(2)$ up to a non-commutative factor in the case of a single D2-brane and a partition $\{N_1, \dots, N_k\}$ of the N dimensional representation of $SU(2)$ in the case of k concentric D2-branes.

The force that the D2 feels from the background is given as

$$F_{D_2}(r) = -\frac{8T_2}{\alpha'q}r^2(r - \pi q\alpha'\mu). \quad (2.32)$$

In fact, we see that it is independent of the configuration of the geometry, for example independent of the warp factor $e^{\frac{2\phi}{3}}$, but it does depend on its own charge q and the flux parameter μ , in accordance with the D0-brane case. We also see that the force becomes repulsive for $r \in (0, \pi q\alpha'\mu)$. This results to an interpolation from a strong, attractive force to a slightly repulsive force after going through the minimum of the potential. In the minimum, we have a D2-brane freely sitting on a position in spacetime. So in this way, we can see that we probe and reveal different parts of the geometry since this picture in the D0-probe was absent.

3 BMN vacua

In this section, after discussing the matrix model vacua and how one can think this as a electrostatic configuration problem following [?], we describe the gravity duals. For complementary material regarding this treatment see [?], [?], [?]. Furthermore, we calculate the force acting on a D0-brane from the geometry.

3.1 Generic analysis

In general, we should look for a gravity solution that preserves half of the supersymmetries⁷ and respects the $SO(3) \times SO(6)$ symmetry. This can be motivated from the specific form of the Hamiltonian

⁷This is because the BMN model has various BPS states, but here in particular we are interested for 1/2-BPS states.

of the matrix model (1.4) since the gravity solution should respect the symmetries therein. The most general solution in type IIA string theory that meets these conditions is [?]

$$ds_{10}^2 = \left(\frac{\ddot{V} - 2\dot{V}}{-V''} \right)^{\frac{1}{2}} \left[-\frac{4\ddot{V}}{\ddot{V} - 2\dot{V}} dt^2 - \frac{2V''}{\dot{V}} (d\rho^2 + dz^2) + 4d\Omega_5^2 + 2\frac{V''\dot{V}}{\Delta} d\Omega_2^2 \right], \quad (3.1)$$

with

$$\begin{aligned} \dot{V} &= \rho \partial_\rho V, \quad \ddot{V} = \rho \partial_\rho(\dot{V}), \quad V' = \partial_z V, \quad V'' = \partial_z^2 V, \\ \Delta &= \left(\ddot{V} - 2\dot{V} \right) V'' - \left(\dot{V}' \right)^2, \end{aligned}$$

and the dilaton and NS-NS and R-R forms are given as

$$\begin{aligned} e^{4\phi} &= \frac{4(2\dot{V} - \ddot{V})^3}{V''\dot{V}^2\Delta^2}, \quad B_2 = 2 \left(\frac{\dot{V}\dot{V}'}{\Delta} + \rho \right) d\Omega_2 \\ A_1 &= \frac{2\dot{V}\dot{V}'}{2\dot{V} - \ddot{V}} dt, \quad A_3 = -\frac{4\dot{V}^2 V''}{\Delta} dt \wedge d\Omega_2. \end{aligned} \quad (3.2)$$

The function $V(z, \rho)$ appearing in (3.1) can be interpreted as a potential that sources different brane (NS5, D2, D0) configurations, and depending on the specific form of this function one realizes different gravity solutions, which correspond to different vacua in the matrix model side. We are interested in the bosonic sector of BMN because in that case we can write a metric ansatz that respects the symmetries. In the BMN case, we see from (1.4) that these symmetries are $R \times SO(3) \times SO(6)$. The first of these implies that there is a Killing vector associated to translations of the time coordinate. The second, implies that we can imagine a sector of the spacetime to consist of a S^2 sphere⁸ while the third the existence of a S^5 sphere, on which the bosonic generators act. Thus one can analyse solutions in eleven dimensions by using only three variables, x_1, x_2, y which satisfy the Toda equation in three dimensions [?],[?]

$$(\partial_{x_1}^2 + \partial_{x_2}^2) V + \partial_y^2 V = 0. \quad (3.3)$$

Here y is used as a combination of the radii of S^2 and S^5 respectively, $y = r_2 r_5^2$. Exploiting the translational invariance under one coordinate (x_1), and after changing variables $(x_2, y) \mapsto (\rho, z)$ one can recast this equation to the Laplace equation in a three dimensional space [?]

$$\frac{1}{\rho} \partial_\rho (\rho \partial_\rho V) + \partial_z^2 V = 0. \quad (3.4)$$

We note that this three-dimensional space is not a sub-part of the ten dimensional one but rather maps to an electrostatic problem in a three dimensional space, where all gravity solutions are classified by ρ, z . Here, both ρ and z are radial coordinates, and hence one can actually think of them as two different scales of the theory. The former captures the size of fivebranes and the latter the positions of the conducting disks, which is actually being set by the five-branes. If one, naively tries to find a gravity solution by just plugging in a solution of (3.4) into the metric (3.1) will end up with singular gravity solutions. To avoid this, one has to impose boundary conditions at $y = 0$ such that $V(y = 0) = 0$.

⁸Actually it is more appropriate to say that we have non-commutative sphere since the coordinates that are assigned to $SU(2)$ generators do not commute.

This condition corresponds to regimes with vanishing radii of S^2 or S^5 . Indeed S^2 shrinks for constant z in the z, ρ space (see figure 3) and S^5 shrinks between two nearby disks in the $\rho = 0$ line.

In the case of the BMN model, the asymptotic expansion is decomposed in two parts which the first of these represents the electrostatic potential and the second one the specific electrostatic configuration imposed by the boundary conditions. In other words, this means that one can specify the solution but not necessary to which vacuum corresponds, since in principle there will be vacua with the same boundary conditions. The first non-trivial potential is [?]

$$V \approx \rho^2 z - \frac{2}{3} z^3 + \mathcal{P} \frac{z}{(\rho^2 + z^2)^{\frac{3}{2}}}, \quad (3.5)$$

where $\mathcal{P}$ is a short of dipole sub-leading correction, produced by the conducting disks and their images with respect to the infinite conducting plane at $z = 0$. Its value is given by

$$\mathcal{P} = 2 \sum_i z_i Q_i \sim N, \quad (3.6)$$

where i runs over the various disks, Q_i is the charge of the i -th disk and N is the total number of D0-branes.⁹ The number of D0-branes can be extracted from the particular configurations of the disks. Suppose for example that there is a finite set of conducting disks at positions $z_i = \frac{\pi}{2} N_5^i$, with radii R_i and charge $Q_i = \frac{\pi^2}{8} N_2^i$. Then the disks are sourced by the NS5-branes in the background, with the charge Q_i being associated to the D2-brane charges. In this sense, one can write the number of D0-branes as

$$N = \sum_i \left(\sum_{j < i} N_5^j \right) N_2^i. \quad (3.7)$$

Furthermore, N_2^j is related to the copies of dimension $N_5^j = (2j+1)$ irreducible representations, giving a geometric realization of the situation of the matrix model. These, correspond to concentric spheres and due to the non-commutativity are actually concentric fuzzy spheres. We will not analyse these fuzzy geometries in this paper, since our interest is for the trivial vacuum.

To get the trivial vacuum metric one has to perform a two steps procedure. The first step is to plug in (3.5) into the metric (3.1) and expand at large ρ and z . Then change coordinates as

$$r^2 = 16 (\rho^2 + z^2), \quad (3.8)$$

and in the large r limit the metric approximates the near horizon geometry of N coincident D0-branes [?]

$$ds^2 \approx -H(r)^{-\frac{1}{2}} dt^2 + H(r)^{\frac{1}{2}} (dr^2 + r^2 d\Omega_8^2) + \dots, \quad H(r) \approx \frac{256 \cdot 15 \mathcal{P}}{r^7}. \quad (3.9)$$

However the large r region is a bit subtle. Let us try to clarify some possible issues. The supergravity description is valid when $g_{\text{eff}} = \frac{\lambda}{\mu^3} \gg 1$ while for $g_{\text{eff}} \ll 1$ we can trust perturbative calculations in matrix model. The first case corresponds to a region between the black hole (i.e near the core of the geometry in figure 4) and $r \sim \lambda^{1/3}$, while the second case is near the boundary. However, near the boundary the gravity solution is not valid since the curvature is high (see next

⁹Here we make contact with the introductory remarks, where we stated that all the contents can be reconstructed from D0-branes.

subsection). Hence we are in a region when r is large but nevertheless not near the boundary. It is large with respect of the core of the geometry and a good estimate would be $r \lesssim \lambda^{1/3}$. Here we can imagine the following situation, we suppose that we want to study the system in a specific energy scale (i.e fixed λ), where we can trust the supergravity solution. This amounts to

$$r \gg 1 \quad \text{and} \quad g_{\text{eff}} = \frac{\lambda}{\mu^3} \gg 1, \quad \lambda = \text{fixed}, \Rightarrow \mu \rightarrow 0. \quad (3.10)$$

Thus, to leading order we recover the case of BFSS model and this is what the metric (3.9) tells us about.

3.2 Trivial vacuum

From the gauge theory side, it seems that the easiest classical vacuum, is the one that all matrices vanish trivially, i.e $X^i = 0 = X^a$, with $i = 1, 2, 3$, $a = 4, \dots, 9$. This corresponds to the situation where we have N copies of the trivial ($j = 0$) representation of $SU(2)$, and in the electrostatic description this translates to $N_2 = N$ and $N_5 = 1$, i.e a single disk at a distance close to (of order one unit) the infinite conducting disk at $z = 0$ (see figure 3). This vacuum is believed to correspond to a single fivebrane in the gravity side [?], and the gravitational approximation is believed to not be very good near the fivebrane [?], [?]. Nevertheless, we perform an expansion in the metric in the large ρ limit while keeping z of order one. Thus the trivial vacuum limit corresponds to

$$r^2 = 16 (\rho^2 + z^2) \gg 1, \quad \rho \gg 1, \quad z \sim \mathcal{O}(1). \quad (3.11)$$

3.2.1 Zero temperature

We will start applying this to zero temperature first and then move to finite temperatures in the next subsection. The recipe indicates that we firstly plug (3.5) into (3.1) and then taking the trivial vacuum, we get a metric of the form

$$ds^2 = -H(r)^{-\frac{1}{2}} dt^2 + 16H(r)^{\frac{1}{2}} (dz^2 + z^2 d\Omega_2^2) + H(r)^{\frac{1}{2}} (dr^2 + r^2 d\Omega_5^2), \quad (3.12)$$

$$H(r) := \frac{256 \cdot 15\mathcal{P}}{r^7}, \quad (3.13)$$

$$e^\phi = \left(\frac{2^{12} 15\mathcal{P}}{r^7} \right)^{\frac{3}{4}}, \quad (3.14)$$

$$A_1(r) = -\frac{r^7}{2^{11} 15\mathcal{P}} - \frac{7z^2 r^5}{2^8 15\mathcal{P}} - \frac{7z^4 r^3}{192\mathcal{P}} + \frac{r^2}{10} - \frac{7z^6 r}{24\mathcal{P}} - \frac{12z^2}{5}, \quad (3.15)$$

We have kept only leading terms both in the expression of dilaton and the one form. We have kept explicitly the z dependence but one should remember that is of order one. The next correction to the dilaton is of order $\mathcal{O}(r^{-29/4})$ and in the large r limit we neglect it. In the same limit, the contribution of the R-R one form is significant. Furthermore, we see that in this limit and for same energy scale (or the same radius r), the effective curvature of the five-sphere, which in string units is the inverse of $\frac{\mathcal{R}_5^2}{\alpha'} = H(r)^{\frac{1}{2}} r^2$, is much smaller than the one of the two-sphere, which is the inverse of $\frac{\mathcal{R}_2^2}{\alpha'} = H(r)^{\frac{1}{2}} z^2$. This means that curvature corrections coming from the two-sphere part of the metric may become

important when compared with the ones of the five-sphere [?] ¹⁰. We also note that the validity of the supergravity solution is when the dilaton is not too large and when the effective curvature of the five-sphere is not too big. Recalling $\mathcal{P} \sim N$ this translates to the range

$$N^{\frac{1}{7}} \ll r \ll N^{\frac{1}{3}}. \quad (3.16)$$

The Ricci scalar of (3.12), can be computed and it is given by

$$\mathcal{R} = -\frac{49\sqrt{\frac{3}{5}}}{64} \left(\frac{r^3}{\mathcal{P}} \right)^{\frac{1}{2}}. \quad (3.17)$$

We see that in the regime where the supergravity description for the trivial vacuum is a good approximation, the spacetime is almost flat provided that N is large.

Lastly, before moving to finite temperature, it worths mentioning that after a coordinate rescaling and transformation, namely

$$z \rightarrow \frac{\zeta}{4}, \quad R \rightarrow \frac{2}{5} \frac{r_0^{5/2}}{r^{5/2}}, \quad r_0^7 := 256 \cdot 15\mathcal{P}, \quad t \rightarrow r_0 t \quad (3.18)$$

the metric (3.12) becomes

$$ds^2 = \left(\frac{2}{5} \right)^{7/5} \frac{r_0^2}{R^{7/5}} \left[-dt^2 + dR^2 + \left(\frac{5}{2} \right)^2 R^2 d\Omega_5^2 \right] + \left(\frac{2}{5} \right)^{-7/5} R^{7/5} (d\zeta^2 + \zeta^2 d\Omega_2^2). \quad (3.19)$$

Therefore, the near horizon limit of the vacuum solution is conformally equivalent to $AdS_2 \times S^5 \times S^2$ if we keep r_0 fixed which means to keep $\mathcal{P}$ fixed. It is obvious (from the limits of validity) that the "size" of the AdS_2 part of the metric is much bigger than the S^2, S^5 , since the trivial vacuum corresponds to large $r^2 = 16(\rho^2 + z^2)$, with ρ large and z of order one. We note however that in the range of validity the S^5 part is bigger than the S^2 part of the metric.

3.2.2 Force at finite temperature

At finite temperature the BMN model has 1/2- BPS states [?], [?] and we should expect the force to be present even if we consider fermions. Eventually, also we would like to make some contact with the quantum mechanics side. The quantum mechanics however is a subsector of the SYM theory. So let us be a bit more illustrative here. The parameters that control the vacua of the BMN, are the number¹¹ of D2-branes N_2 , the number of D5-branes N_5 , and the number of D0-branes N_0 , whose relation is given as [?]

$$N_2 = \frac{N_0}{N_5}. \quad (3.20)$$

The trivial vacuum is dictated by $N_5 = 1, N_2 = N_0 = N$, with N being the copies of the trivial ($j = 0$) representation. This particular configuration is given in figure 3. Then to understand the situation in

¹⁰Both r and z here are energy scales, and this condition reflects the fact that we push the regime of validity of the metric to its boundary. In [?], and based on [?] was argued that the breaking of Einstein supergravity starts at scales $\lambda^{1/3}$.

¹¹To be more precise it is the charge of the various D-branes.

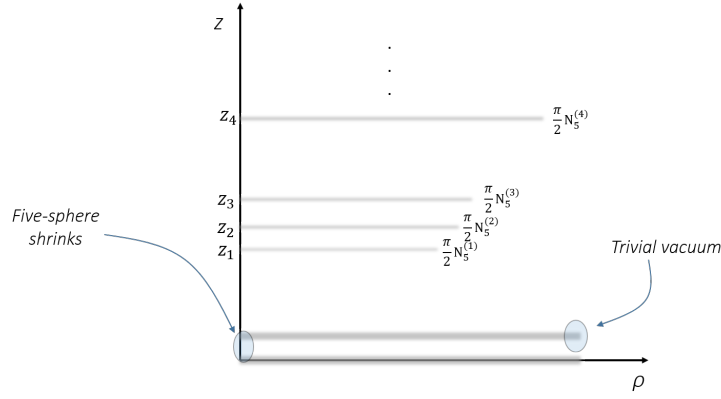


Figure 3: A cartoon of the $z-\rho$ plane. Constant z lines indicate that the two-sphere shrinks while the five-sphere shrinks in the region between nearby disks near $\rho = 0$. We have a conducting disk at $z = 0$ which extends to infinity in the ρ direction.

the gauge theory side, we need to convert the gravity data to the gauge data. By direct comparison with [?], we find the following [?]¹²

$$\rho_0^4 = \frac{\pi^4}{2} g_{YM}^2 N \quad , \quad d = \frac{\pi}{2} N_5 = \frac{\pi}{2} \quad , \quad \mathcal{Q} = \frac{\rho_0^4}{8} \quad , \quad \mathcal{P} = 2d \mathcal{Q} = \frac{\pi^5 g_{YM}^2 N}{16}. \quad (3.21)$$

Here, ρ_0 is the value of the coordinate ρ on which the tip of the conducting disk lies (see figure 3), d is the charge of N_5 which is also the distance between the two conducting disks. Before we continue, the above data reflect the repeated statement, that the size of the fivesphere is much larger than the one of S^2 , into the radius of the fivesphere [?]

$$\frac{R_{S^5}^2}{\alpha'} = 4\pi\rho_0 = 4\pi \left(\frac{\lambda}{2\mu^3} \right)^{\frac{1}{4}}. \quad (3.22)$$

We investigate the gravity solution at finite temperature T in this vacuum. We can write the finite T black brane solution as

$$\begin{aligned} \frac{ds_T^2}{\alpha'} &= -H(r)^{-\frac{1}{2}} f(r) dt^2 + 16H(r)^{\frac{1}{2}} (dz^2 + z^2 d\Omega_2^2) + H(r)^{\frac{1}{2}} \left(\frac{dr^2}{f(r)} + r^2 d\Omega_5^2 \right), \\ H(r) &= \frac{\lambda d_0}{r^7} \quad , \quad e^\phi = 2^6 \left(\frac{\lambda 15 \pi^5}{r^7} \right)^{\frac{3}{4}} \quad , \quad f(r) = 1 - \frac{r_0^7}{r^7}, \\ A_1(r) &= -\frac{r^7}{1920\pi^5 \lambda} - \frac{7r^5}{960\pi^3 \lambda} - \frac{7r^3}{192\pi \lambda} + \frac{r^2}{10} - \frac{7\pi r}{96\lambda} - \frac{3\pi^2}{5}, \\ d_0 &\equiv 240\pi^5 \quad , \quad \lambda \equiv g_{YM}^2 N \quad , \quad g_s = 4\pi^2 l_s^3 \frac{\lambda}{N} \quad , \quad G_{10} = g_s^2 l_s^8, \end{aligned} \quad (3.23)$$

and the horizon is located at $r = r_0$, with r_0 representing the thermal energy scale. The horizon is dictated only from the radius r , but we have implicitly assigned $r^7 := r_5^5 \cdot z^2$, where r_5, z the radii of the 5-sphere and 2-sphere respectively. In other words, we have merged the two scales into one. We can proceed even more and pick a scale for the S^2 part such that we measure everything at this fixed

¹²We suppress μ dependence here by setting $\mu = 1$ but we can restore it by using $g_{\text{eff}} = \frac{\lambda}{\mu^3}$ and by dimensional analysis, e.g $[\alpha'] = 2$, $[\mu] = 1$, $[\lambda] = 3$ in units of energy.

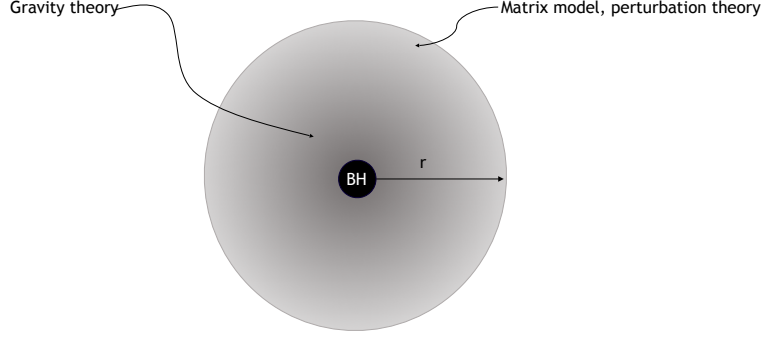


Figure 4: A pictorial representation of the regime of gravity and the regime of matrix model. In this way one can approximate different regimes with the help of the radial parameter r , the deformation parameter μ and the effective coupling $g_{\text{eff}} = \frac{\lambda}{\mu^3}$.

scale. We will choose this to be $z = \frac{\pi}{2}r_2$ and this was already implemented in the expressions of (3.23) by setting $r_2 = 1$, but not in the metric itself. This choice reflects the trivial vacuum limit discussed above and will be used when we calculate the force.

Demanding the curvature of the spheres to be not too large we get the condition

$$r \lesssim \lambda^{1/3} \quad \text{both from } S^5 \text{ and } S^2. \quad (3.24)$$

There is also one more constraint coming by demanding that the dilaton is not too large on the horizon which results in

$$\lambda\beta^3 \lesssim N^{10/7}, \quad (3.25)$$

and for sufficiently large N with respect to the combination $\lambda\beta^3$ the last condition is automatically satisfied.

From this metric one can extract the temperature via the surface gravity

$$T = \frac{7r_0^{5/2}}{4\pi\sqrt{\lambda d_0}} \quad (3.26)$$

The calculation of the force uses a variation of section 2,

$$V_{D_0}(r) = T_0 \left[e^{-\phi} H(r)^{-\frac{1}{4}} f(r)^{\frac{1}{2}} - A_1(r) \right]. \quad (3.27)$$

with the data of the finite temperature gravity solution (3.23). We expect qualitatively similar behaviour as in the zero temperature system because of BPS states, and evaluating indeed we see the behavior in figure 5. Here we show the dependence of everything in terms of λ to make conduct

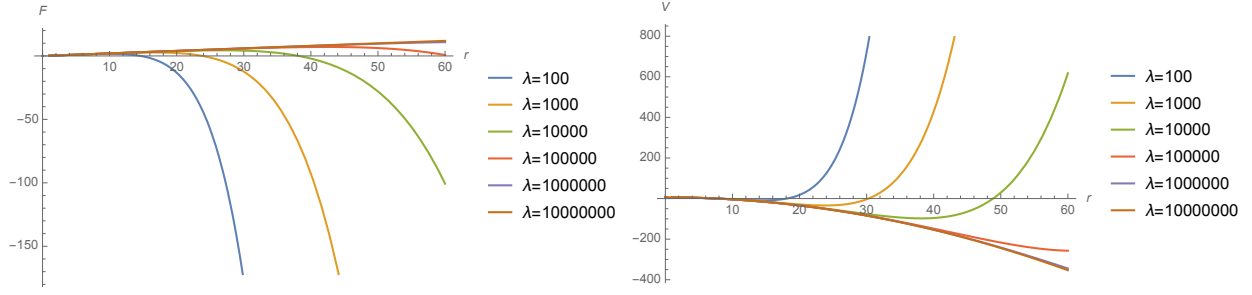


Figure 5: Finite temperature force (left) and potential (right) with respect to the radial coordinate r (or equivalently temperature T). We have set for the horizon $r_0 = 1$ and $g_{YM} = 1$.

with the existing literature. One, can understand that there is also a hidden μ dependence if we use the dimensionless coupling $g_{\text{eff}} = \frac{\lambda}{\mu^3}$.

What is different here in contrast to the BFSS case, is the fact that repulsive phenomena can occur even classically and far away from the horizon if λ is sufficiently large. This is also the regime where the gravity description is valid, and we can understand this repulsive phenomenon as the R-R charge of the D0-branes. Moreover, it is known for the BMN model that there might be a tunneling effect between "nearby" vacua [?], where the probability to tunnel is given by the exponential $e^{-\frac{N}{\lambda}}$ since it is mediated by instantons. N is the number of the degrees of freedom (i.e D0-branes) and λ the t'Hooft coupling. As a result in the regime we are interested in, the tunneling between different vacua is exponentially suppressed.

The main lesson is that the behaviour of the D0-probe reported here is at least *qualitatively* similar to quantum corrected black 0-brane geometries [?] which were verified by matrix simulations [?]. The analysis performed here is purely classical and furthermore is λ independent. This can be seen from the fact that the curves in figure 5 join going towards $r \rightarrow 0$ and the steep regions are outside of the regime of validity. One can imagine fixing λ and taking $\mu \rightarrow 0$ such that the condition of supergravity $g_{\text{err}} \gg 1$ is satisfied. Then, in principle this behaviour should be visible in matrix simulations along the lines of [?].

We also note that the geometry which is believed to correspond to BFSS model is recovered when we take $\mu \rightarrow 0$. At first sight this might not be obvious but we observe the following. We always want the gravity description to be valid which results in $g_{\text{eff}} \gg 1$ and we also want $r \gg 1$ such that we are away from the singularity. If we fix λ which amounts to fix the energy scale of the system, we have that

$$g_{\text{eff}} = \frac{\lambda}{\mu^3} \gg 1, \quad \lambda = \text{fixed} \quad \Rightarrow \quad \mu \rightarrow 0. \quad (3.28)$$

Then for fixed $\lambda \gg \mu^3$ with $\mu \rightarrow 0$, the dominant contribution in the R-R one-form comes from the r^7 term for large r . This is also the case for BFSS model where the R-R form is given in terms of r^7 [?], [?]. As we saw, the same limit is what constructs the metric (3.9) to leading order, which is the metric for the gravity dual of the BFSS.

4 Synopsis

We have demonstrated gravitational behaviors of D-brane probes (D0-and D2-branes) for symmetry reduced pp-wave backgrounds. In particular for the D0-probe we have also studied its behavior for a geometry which corresponds to the trivial vacuum of BMN. We have seen a repulsive behavior as we approach the core in the extremal case and the horizon in the non-extremal case. This most likely indicates that there is a mechanism that gives rise to repulsive forces as one approaches the core [?]. We have demonstrated, that this mechanism can occur both at classical level and also far away from the core for the BMN model. One possible way to understand this intuitively is because as an observer (here a D0-brane) approaches the core, firstly meets the S^5 part of the horizon surrounding the core and then the S^2 . This particular probe measures everything in terms of the $SU(2)$ part of the geometry since we have fixed the scale z that controls the size of S^2 . This is a choice based on observations made in subsection 2.1. There we have showed that a single D0-brane is agnostic for the existence of $SU(2)$ while in the next subsection we have demonstrated that a D2-brane is aware for the $SU(2)$ part. Now the charges of S^2 and S^5 consisting from D0-branes carry one forms with which the probe brane couples naturally. However, having the same charge the one form $A_1(r)$ causes repulsion. Moreover, this is something that classically does not appear in the BFSS model but rather when one considers quantum corrections [?]. It is natural to expect that this deformed geometry of BMN model modifies slightly the behaviour of the D0-probe even classically since they are in principle different backgrounds.

Nevertheless, the regime where we can trust supergravity depends on g_{eff} and hence for large g_{eff} one is already far away from the singularity. Sufficiently far away the D0-probe will be at the minimum of the potential. This follows from the analysis done in the main text and from the principle of least action and it holds only in the trivial vacuum geometry. Further studies need to be carried out to understand the situation of the non-trivial vacua which from geometric point of view are significantly more complicated but the intuitive understanding is as concentric fuzzy spheres. However, one significant caveat of the construction is that we did not check explicitly that the vacuum geometries (3.12) and (3.23) are solutions of Einstein's equations, but it is rather based on the spirit of [?].

Understanding these bulk descriptions will be beneficial to extract meaningful physics when one also compares with the matrix model side both in analytic, and nowadays, in numerical results.

Acknowledgments